

Java Merge Sort

Complete Java program for Merge Sort in ascending order.

```
import java.util.*;

public class Main {

    static void mergeSort(int[] arr, int left, int right) {
        if (left < right) {
            int mid = (left + right) / 2;

            mergeSort(arr, left, mid);
            mergeSort(arr, mid + 1, right);

            merge(arr, left, mid, right);
        }
    }

    static void merge(int[] arr, int left, int mid, int right) {
        int i = left;
        int j = mid + 1;
        int k = 0;

        int[] temp = new int[right - left + 1];

        while (i <= mid && j <= right) {
            if (arr[i] <= arr[j]) {
                temp[k++] = arr[i++];
            } else {
                temp[k++] = arr[j++];
            }
        }

        while (i <= mid) {
            temp[k++] = arr[i++];
        }

        while (j <= right) {
            temp[k++] = arr[j++];
        }

        for (i = left, k = 0; i <= right; i++, k++) {
            arr[i] = temp[k];
        }
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        mergeSort(arr, 0, n - 1);

        for (int x : arr) {
            System.out.print(x + " ");
        }
    }
}
```

Example Input

```
5
38 27 43 3 9
```

Example Output

```
3 9 27 38 43
```